Attack of the killer Pathogens....

The Immune System is a complex group of organs and cells that defends the body against infection or disease. Communicable diseases include those resulting from the invasion of foreign particles into the body and multiplying. These disease causing organisms are called PATHOGENS, and some are CONTAGIOUS- which means that they can be passed on by direct contact from somebody who has the disease. Pathogens can also be transmitted by VECTORS, which are hosts of the disease and examples include mosquitoes.

There are 4 examples of pathogens and these include: Bacteria, Viruses, Fungi and Animal Parasites. Each pathogen type has different characteristics which enable it to invade and multiply in the body.

The spread of pathogens from one person to another can occur in a number of ways.

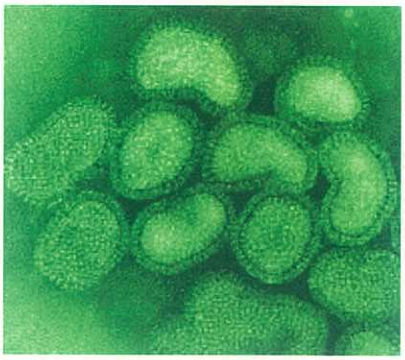
1. Transmission by c _____ - spread by with directly touching an infected person or indirectly by touching something that has been touched by an affected person.
Examples include: Skin infections and Venereal Diseases.

2. Transmission by droplets-occur when droplets of moisture are let out during exhalation, talking, sneezing or coughing.
Examples include: Colds and Influenza

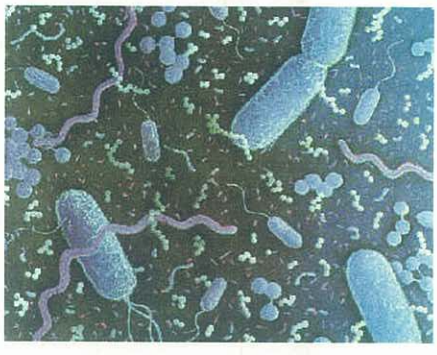
3. Ingestion-by eating or drinking contaminated particles.
Examples include: Salmonella Food Poisoning

4. Airborne Transmission- through exhalation and coughing many viruses are able to be transmitted.
Eg: SARS

5. Vectors- Animals such as fleas, ticks and mosquitoes transferring pathogens.
Examples include: Mosquitoes and Malaria



Viruses



Bacteria Cells

Catch my disease....

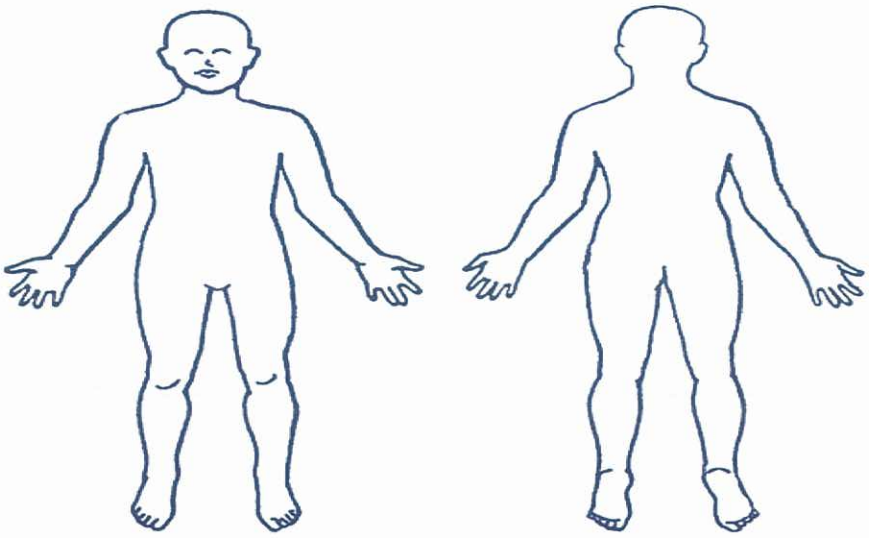
Using computer resources, you are to research **5 diseases** caused by each pathogen type and record your information in the table below.

NAME OF PATHOGEN ORDISEASE	Type of organism causing it.	TRANSMISSION –how do you get it? Who is likely to get it?	DESCRIPTION OF THE DISEASE - symptoms	TREATMENT AND OUTLOOK
	BACTERIA			
	VIRUS			

	FUNGI			
	GENETIC			
	ANIMAL PARASITE			

External Boarder Security!!!!

Using the body below, label in the external body defences described below.



1. **TONSILS**: - are made of lymphoid tissue, located at the entrances to the throat, they act as barriers to bacteria & viruses.

2. **TEAR DUCTS**: - located above the outer corner of each eye. They secret tears, which allow for dirt to be washed out of the eyes. Tears also contain Lysozyme, a substance that combats bacteria.

3. **SALIVARY GLANDS**: - are found in the cheeks and under the tongue. The salvia they secrete contains substances, which help to resist infection.

4. **MUCOUS MEMBRANES**: - is a soft moist tissue lining the body openings and passages, such as the nose & the throat. It secretes mucus, a sticky fluid, which provides protection against some bacteria. In the nasal cavity the mucous membrane also traps germs and dust.

5. **LYMPH NODES**: - Are small lumpy structures scattered along the course of the lymphatic system. The neck, armpits and groins have the greatest numbers. The nodes also produce white blood cells, which attack foreign matter.

6. **SPLEEN:** - which has lymphoid tissue, helps to form white blood cells and remove used ones.

7. **STOMACH**: - has digestive juices including hydrochloric acid which has a pH of 2, which helps to sterilize the food intake.

10. **LYMPHATIC SYSTEM**: - The lymph stream contains white blood cells, which helps to stop the spread of infection by producing antibodies.

11. **SKIN**: - provides the body's main protection against disease. Because it is dry, slightly acidic and impervious invaders rarely invade the skin, except in the case of the skin being broken.

Keeping Australia disease free....

People that arrive in Australia from other countries must be able to show evidence of their vaccinations against certain diseases, or they may be subject to medical examinations on arrival.

1. Why is this so important?

2. Define the following terms:
ENDEMIC-

EPIDEMIC-

PANDEMIC-

3. What are Notifiable Diseases? Why are they necessary?

4. Give 5 examples of notifiable diseases in Australia.

5. Leprosy is endemic to the Northern Territory of Australia where it is found mainly among Aborigines. It isn't very contagious but is found in unhygienic, overcrowded conditions. What could be done to reduce the incidence of leprosy in Australia?

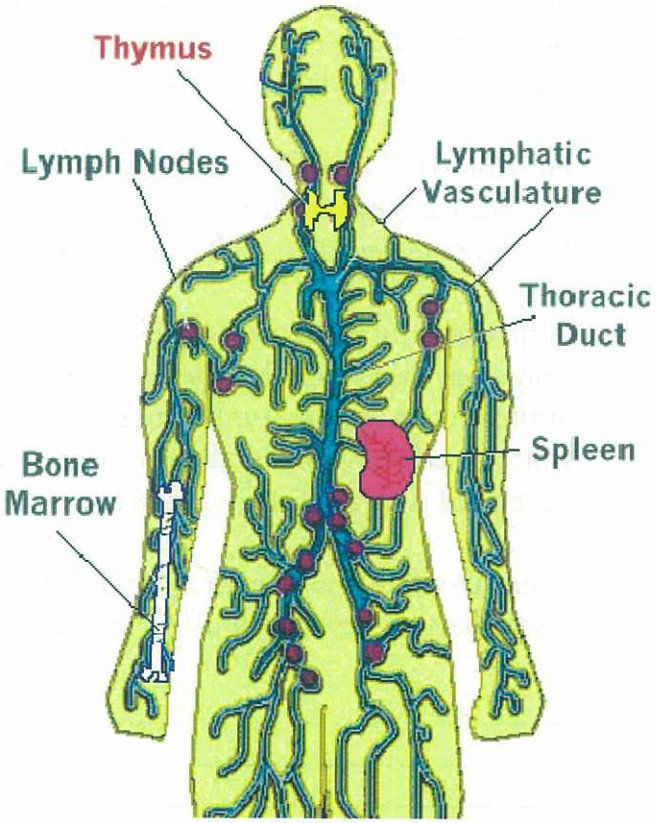
6. "Economically, the virus causing the common cold is the most important of the viruses that cause disease in humans" What do you think would be the economic importance of the cold virus?

7. When antibiotics are prescribed as treatment for bacterial infections, patients are instructed to complete the whole course of tablets until they are finished. Suggest why this is important even if you feel you have recovered before all the tablets are finished.



Internal Boarder Security!!

The Lymphatic system is the body's internal defence system against disease, and comes into action when the external defence system fails.

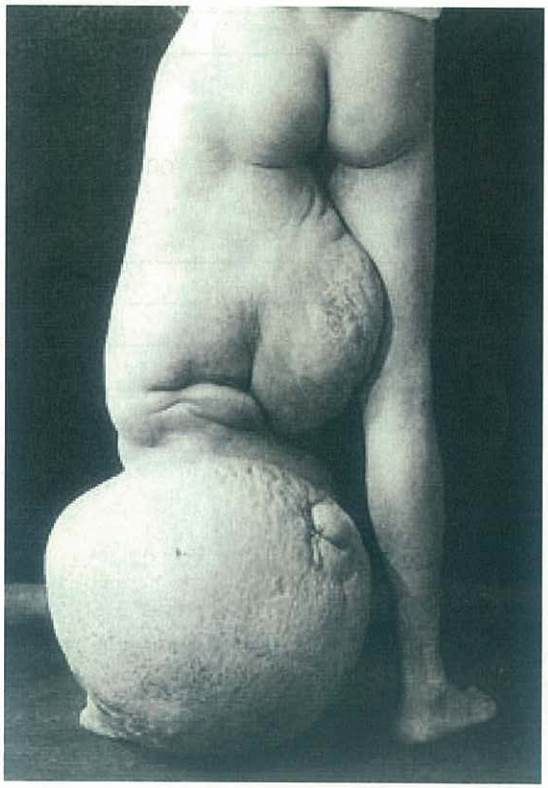


The lymphatic system defends the body from foreign invasion by disease causing agents such as viruses, bacteria, or fungi. The lymphatic system consists macroscopically of: The bone marrow, thymus gland, tonsils, spleen, appendix and a few other organs. The lymph system contains a network of vessels that assists in circulating body fluids. These vessels transport excess fluids away from the spaces between cells in body tissue and return it to the bloodstream. Lymphatic vessels prevent the backflow of the lymph fluid. Lymph Nodes are specialised organs which filter out destroyed pathogens.

Common problems

Some common problems of the lymphatic system include:

- **Glandular Fever** - symptoms include tender lymph nodes
- **Hodgkin's Lymphoma** - a type of cancer of the lymphatic system
- **Oedema** - swelling caused by too much fluid in the tissues
- **Tonsillitis** - infection of the tonsils in the throat.
- **Appendicitis**



I need a prescription!!!!

At times the Immune System of human body is unable to deal with invading micro-organisms quickly enough. These micro-organisms multiply in the body and symptoms of the disease can occur.

ANTIBIOTICS are chemicals that are used to kill infecting **bacteria**. For antibiotics to be effective they must only be toxic to the pathogen and not to the cells of the body. For example Penicillin is effective in treating bacterial infections such as tonsillitis.

Although antibiotics are available for most bacterial and fungal diseases, there are only a few drugs available for the treatment of viral diseases. It is difficult to find antibiotics to treat a virus because when it invades a cell, it takes over and replicates, which interferes with metabolism of the host cell. A few antiviral drugs have been developed such as AZT for the treatment of HIV and AIDS.

There are a large number of antibiotics now available, why is it necessary for researchers to look continually for new antibiotics?

How are antibiotics made?



The Killer Virus of the 21st Century

What is HIV?

HIV stands for Human Immunodeficiency Virus. It is the virus that causes AIDS. A member of a group of viruses called retroviruses, HIV infects human cells and uses the energy and nutrients provided by those cells to grow and reproduce.

What is AIDS?

AIDS stands for Acquired Immune Deficiency Syndrome. It is a disease in which the body's immune system breaks down and is unable to fight off infections, known as "opportunistic infections," and other illnesses that take advantage of a weakened immune system.

When a person is infected with HIV, the virus enters the body and lives and multiplies primarily in the white blood cells. These are immune cells that normally protect us from disease. HIV infection results in the progressive loss of a specific type of immune cells called T-helper cells. As the virus grows, it damages or kills these and other cells, weakening the immune system and leaving the person vulnerable to various opportunistic infections and other illnesses ranging from pneumonia to cancer.

How quickly do people infected with HIV develop AIDS?

In some people, the T-cell decline and opportunistic infections that signal AIDS develop soon after infection with HIV. But most people do not develop symptoms for 10 to 12 years, and a few remain symptom-free for much longer.

How many people are affected by HIV/AIDS?

The Joint United Nations Programme on HIV/AIDS (UNAIDS) estimates that there are now 40 million people living with HIV or AIDS worldwide. Most of them do not know they carry HIV and may be spreading the virus to others. In the U.S., ~ one million people have HIV or AIDS, and 40,000 Americans become newly infected with HIV each year. It is estimated that a quarter of all people with HIV in the U.S. do not know they are carrying the virus.

Since the beginning of the epidemic, AIDS has killed more than 25 million people worldwide, including more than 500,000 Americans. AIDS has replaced malaria and tuberculosis as the world's deadliest infectious disease among adults and is the fourth leading cause of death worldwide. Fifteen million children have been orphaned by the epidemic.

Did you Know?

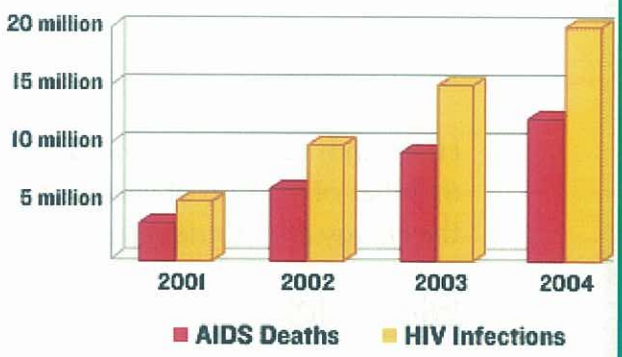
Since the first HIV case 20 years ago, over **60 million** persons have been infected, and over **20 million** have already died from AIDS.

In Africa, there are an estimated **11,000** new infections per day, and during 2001 approximately **2.3 million** Africans will die from HIV/AIDS.

In several Caribbean countries, HIV/AIDS has become a leading **cause of death**. Haiti and the Bahamas are especially hard hit.

Eastern Europe – especially the Russian Federation – continues to experience the **fastest-growing epidemic**.

Projected Cumulative Global AIDS Death and New HIV Infections 2001-2004



SOME STATISTICS

Currently, there are approximately **14 million** children orphaned by HIV/AIDS, with a projection of 40 million children by 2010 if no action is taken.

With **each minute** that passes, another African child dies of AIDS.

About **one-third** of those currently living with HIV/AIDS are aged 15-24. Most of them do not know they carry the virus. Many millions more know nothing or too little about HIV to protect themselves against it.

In sub-Saharan Africa, the economic hardships of the past two decades have left three-quarters of the continent's people surviving on less than **US $2** a day.

7 million farm workers have died from AIDS-related causes since 1985 and **16 million** more are expected to die in the next 20 years..

In 1999 alone, an estimated **860,000** children lost their teachers to AIDS in sub-Saharan Africa.

Were it not for HIV/AIDS, average life expectancy in sub-Saharan Africa would be about 62 years; instead, it is now **47 years**.



Live and let live

1. In what ways does HIV differ from other virus's that cause disease in humans?

2. List the ways HIV transmitted from person to person

3. What causes death in patients who develop AIDS following HIV infection?

4. Why has AIDS been able to spread so rapidly throughout Australia?

5. What are the current treatments available for HIV infection?



Continuity of Human Life

The survival of the species relies upon reproduction, and to continue our species, some members must produce new individuals to replace those that die. Humans reproduce sexually- a process involving the fusing of male and female gametes- sex cells. These gametes are produced in the gonads- the testes and the ovaries. Reproduction requires the most sophisticated workings of the human body, relying on its specialised structures, timing and the efforts and glands and hormones. Like Yin and Yang, reproduction involves the fusing of 2halves into 1 whole- that being male and female. The male body is designed to allow the manufacture, storage and transfer of male sex cells- the sperm, while the female is designed to manufacture, store and release ova- the eggs.

The Male Reproductive System

Learn the male reproductive structures shown below.

